

LAB 3: SCANNING AND ENUMERATION

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Scanning and enumeration are part of the information-gathering process, whereby the penetration tester obtains a list of available hosts and investigates the services available on each host. This is achieved via a combination of network scanning via a ping sweep and port scanning approaches.

1. Create a NAT Network in your virtual machine to simulate the internal network of a target organization. Use the table below to record the IP address of each VM host.

Answer:

Created a new a NAT network (10.0.2.0/24)

Hostname	OS	IP Address
MSEdgeWIN10	Windows 10	10.0.2.15
Metasploitable 2	Linux (Ubuntu)	10.0.2.3

Windows 10

```
C:\Users\IEUser>systeminfo

Host Name:                MSEdgeWIN10
OS Name:                   Microsoft Windows 10 Enterprise Evaluation
OS Version:               10.0.17763 N/A Build 17763
OS Manufacturer:         Microsoft Corporation
OS Configuration:         Standalone Workstation
OS Build Type:             Multiprocessor Free


C:\Users\IEUser>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::30b0:1664:7729:63f6%7
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1
```

Metasploitable 2

```
msfadmin@metasploitable:~$ uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 G
NU/Linux

msfadmin@metasploitable:~$ ifconfig eth0
eth0      Link encap:Ethernet  HWaddr 08:00:27:bf:43:ad
          inet addr:10.0.2.3  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:febf:43ad/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:117 errors:0 dropped:0 overruns:0 frame:0
          TX packets:115 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:18775 (18.3 KB)  TX bytes:16781 (16.3 KB)
          Base address:0xd020  Memory:f0200000-f0220000
```

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2. Run a ping sweep using the NMAP tool to discover the hosts in the NAT network. Exclude from pinging the Kali Linux (Attacker) machine.

- a. Write/Screen capture the NMAP command below.

Metasploitable 2

```
(pentest@PenTest)-[~]
$ nmap -sn 10.0.2.0/24 --exclude 10.0.2.19
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 11:54 +08
Nmap scan report for 10.0.2.1 (10.0.2.1)
Host is up (0.00043s latency).
MAC Address: 52:55:0A:00:02:01 (Unknown)
Nmap scan report for 10.0.2.2 (10.0.2.2)
Host is up (0.00025s latency).
MAC Address: 08:00:27:00:86:D7 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap scan report for 10.0.2.3 (10.0.2.3)
Host is up (0.00035s latency).
MAC Address: 08:00:27:BF:43:AD (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00030s latency).
MAC Address: 08:00:27:C1:5F:83 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap done: 255 IP addresses (4 hosts up) scanned in 2.01 seconds
```

Windows 10

```
(pentest@PenTest)-[~]
$ nmap -sn 10.0.2.0/24 --exclude 10.0.2.19
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 11:54 +08
Nmap scan report for 10.0.2.1 (10.0.2.1)
Host is up (0.00043s latency).
MAC Address: 52:55:0A:00:02:01 (Unknown)
Nmap scan report for 10.0.2.2 (10.0.2.2)
Host is up (0.00025s latency).
MAC Address: 08:00:27:00:86:D7 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap scan report for 10.0.2.3 (10.0.2.3)
Host is up (0.00035s latency).
MAC Address: 08:00:27:BF:43:AD (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00030s latency).
MAC Address: 08:00:27:C1:5F:83 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap done: 255 IP addresses (4 hosts up) scanned in 2.01 seconds
```

- b. List out all the IP addresses that NMAP have managed to find.
10.0.2.1
10.0.2.2
10.0.2.3 (Metasploitable 2)
10.0.2.15 (Window 10)
- c. Did NMAP manage to scan all the IP addresses listed in the table in Question (1)? If not, why?
Yes. After configuring all VMs to use the NAT network, they were connected to the same virtual network and gateway, allowing Nmap to detect them.
- d. Use the Metasploitable host IP address as the decoy and re-run the ping sweep. Write/screenshot the command below.

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```
(pentest@PenTest)-[~]
$ nmap -D 10.0.2.3 -sn 10.0.2.0/24 --exclude 10.0.2.19
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 12:00 +08
Nmap scan report for 10.0.2.1 (10.0.2.1)
Host is up (0.00022s latency).
MAC Address: 52:55:0A:00:02:01 (Unknown)
Nmap scan report for 10.0.2.2 (10.0.2.2)
Host is up (0.00013s latency).
MAC Address: 08:00:27:00:86:D7 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap scan report for 10.0.2.3 (10.0.2.3)
Host is up (0.00024s latency).
MAC Address: 08:00:27:BF:43:AD (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00026s latency).
MAC Address: 08:00:27:C1:5F:83 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap done: 255 IP addresses (4 hosts up) scanned in 2.00 seconds
```

3. To bypass the firewall, an attacker can opt to conduct an ARP scan to discover live hosts in a network

- a. Run an **active** ARP scan using the netdiscover tool.

- i. Write/Screen capture the command below.

```
(pentest@PenTest)-[~]
$ sudo netdiscover -r 10.0.2.0/24
```

Currently scanning: Finished! | Screen View: Unique Hosts

5 Captured ARP Req/Rep packets, from 4 hosts. Total size: 304

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
10.0.2.1	52:55:0a:00:02:01	1	64	Unknown vendor
10.0.2.2	08:00:27:00:86:d7	2	120	PCS Systemtechnik GmbH
10.0.2.3	08:00:27:bf:43:ad	1	60	PCS Systemtechnik GmbH
10.0.2.15	08:00:27:c1:5f:83	1	60	PCS Systemtechnik GmbH

- ii. List all the hosts that netdiscover managed to discover.

10.0.2.1

10.0.2.2

10.0.2.3 (Metasploitable 2)

10.0.2.15 (Windows 10)

- iii. Did the ARP scan manage to find all the hosts listed in the table in Question (1)? Discuss why?

Yes. ARP scans operate at Layer 2 and are typically not blocked by firewalls. Since ARP is essential for local network communication, hosts must respond to ARP requests, making this method more reliable than ICMP-based scans.

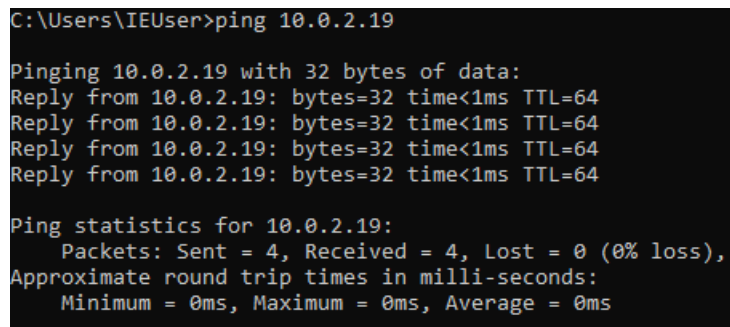
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- b. Run a **passive** ARP scan using the netdiscover tool.
- i. Obtain similar result in Question (3) (a) (ii). Explain the steps.
It get the similar results, able to find all 2 hosts actively pinging my main attacker machine.

Step 1: Add the (-p) flag to listen to ARP request and run cli in kali linux.



Step 2: Generate some traffic from the Windows 10 host.



```
C:\Users\IEUser>ping 10.0.2.19

Pinging 10.0.2.19 with 32 bytes of data:
Reply from 10.0.2.19: bytes=32 time<1ms TTL=64
Reply from 10.0.2.19: bytes=32 time<1ms TTL=64
Reply from 10.0.2.19: bytes=32 time<1ms TTL=64
Reply from 10.0.2.19: bytes=32 time<1ms TTL=64

Ping statistics for 10.0.2.19:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Step 3 : Hosts appear in the results.

Currently scanning: (passive)		Screen View: Unique Hosts		
3 Captured ARP Req/Rep packets, from 2 hosts. Total size: 180				
IP	At MAC Address	Count	Len	MAC Vendor / Hostname
10.0.2.15	08:00:27:c1:5f:83	2	120	PCS Systemtechnik GmbH
10.0.2.2	08:00:27:00:86:d7	1	60	PCS Systemtechnik GmbH

- ii. What is the main advantage of using a passive ARP scan compared to an active ARP scan? And what is its disadvantage?

Advantage:

Passive ARP scanning is stealthier than active scanning because it only listens to existing ARP traffic on the network without sending any probes. This makes it less likely to be detected by intrusion detection systems (IDS) or network monitoring tools.

Disadvantage:

It is slower and less reliable than active scanning because it depends on naturally occurring network traffic. If there is little or no ARP traffic, the scan may not discover all hosts or may take a long time to do so.

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4. Port scanning is used to discover the services that a host provides. TCP scanning is used to discover services running on top of the TCP protocol while UDP scanning is to discover services running on top of UDP protocol.
 - a. Run a TCP Port scanning using the full scan (-sT) approach to discover whether SSH (22) and HTTPS (443) is running on the Metasploitable and Win10 approach.
 - i. Record your result in the table below.

Host	Port Number	Status (Open/Closed/Unknown)
Metasploitable 2	22	Open
	443	Closed
Windows 10	22	Filtered
	43	Filtered

```
(pentest@PenTest)-[~]
$ nmap -sT -p 22,443 10.0.2.3 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 12:21 +08
Nmap scan report for 10.0.2.3 (10.0.2.3)
Host is up (0.00046s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
443/tcp    closed https
MAC Address: 08:00:27:BF:43:AD (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00049s latency).

PORT      STATE SERVICE
22/tcp    filtered ssh
443/tcp    filtered https
MAC Address: 08:00:27:C1:5F:83 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 2 IP addresses (2 hosts up) scanned in 1.42 seconds
```

- ii. Explain your result.

Metasploitable 2 (Port 22 – Open):

The SSH service is running and actively accepting connections.

Metasploitable 2 (Port 443 – Closed):

The HTTPS service is not running on this port, and the host responded with a TCP RST packet, indicating the port is closed.

Windows 10 (Port 22 & 443 – Filtered):

Both ports are blocked by the Windows firewall. No response (SYN-ACK or RST) was received, which is typical for filtered ports the packets are dropped silently.

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- b. Use the **inverse** scanning (-sF) approach to discover whether the same service in (a) is running.

- i. Record your results in the table below.

Host	Port Number	Status (Open/Closed/Unknown)
Metasploitable 2	22	Open filtered
	443	Closed
Windows 10	22	Open filtered
	43	Open filtered

Metasploitable 2

```
(pentest@PenTest)-[~]
$ sudo nmap -sF -p 22,443 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 12:29 +08
Nmap scan report for 10.0.2.3 (10.0.2.3)
Host is up (0.00050s latency).

PORT      STATE      SERVICE
22/tcp    open|filtered ssh
443/tcp    closed     https
MAC Address: 08:00:27:BF:43:AD (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.51 seconds
```

Windows 10

```
(pentest@PenTest)-[~]
$ sudo nmap -sF -p 22,443 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 12:32 +08
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00059s latency).

PORT      STATE      SERVICE
22/tcp    open|filtered ssh
443/tcp    open|filtered https
MAC Address: 08:00:27:C1:5F:83 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.47 seconds
```

- ii. Use the Wireshark software in Kali Linux to study how NMAP conducts inverse scanning. **Screenshot** the Wireshark output and **Explain** how the inverse scanning is conducted based on the Wireshark output.

Metasploitable 2

No.	Time	Source	Destination	Protocol	Length	Info
44	62.407673606	52:55:0a:00:02:01	PCSSystemtec_78:91:...	ARP	64	10.0.2.1 is at 52:55:0a:00:02:01
45	191.192094236	PCSSystemtec_78:91:...	Broadcast	ARP	42	Who has 10.0.2.3? Tell 10.0.2.19
46	191.192578767	PCSSystemtec_bf:43:...	PCSSystemtec_78:91:...	ARP	60	10.0.2.3 is at 08:00:27:bf:43:ad
47	191.307448661	10.0.2.19	192.168.0.1	DNS	81	Standard query 0xfe52 PTR 3.2.0.10.in-addr.arpa
48	191.312050380	192.168.0.1	10.0.2.19	DNS	103	Standard query response 0xfe52 PTR 3.2.0.10.in-addr.arpa PTR 10.0.2.3
49	191.323987068	10.0.2.19	10.0.2.3	TCP	54	34538 → 443 [FIN] Seq=1 Win=1024 Len=0
50	191.324041447	10.0.2.19	10.0.2.3	TCP	54	34538 → 22 [FIN] Seq=1 Win=1024 Len=0
51	191.324399226	10.0.2.3	10.0.2.19	TCP	60	443 → 34538 [RST, ACK] Seq=1 Ack=2 Win=0 Len=0
52	192.426005883	10.0.2.19	10.0.2.3	TCP	54	34540 → 22 [FIN] Seq=1 Win=1024 Len=0
53	196.334047241	PCSSystemtec_bf:43:...	PCSSystemtec_78:91:...	ARP	60	Who has 10.0.2.19? Tell 10.0.2.3
54	196.334081452	PCSSystemtec_78:91:...	PCSSystemtec_bf:43:...	ARP	42	10.0.2.19 is at 08:00:27:78:91:a2
55	196.551534887	PCSSystemtec_78:91:...	52:55:0a:00:02:01	ARP	42	Who has 10.0.2.1? Tell 10.0.2.19
56	196.552078428	52:55:0a:00:02:01	PCSSystemtec_78:91:...	ARP	64	10.0.2.1 is at 52:55:0a:00:02:01

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Windows 10

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.2.19	10.0.2.2	DHCP	327	DHCP Request - Transaction ID 0xfe2a0d38
2	0.008095144	10.0.2.2	10.0.2.19	DHCP	590	DHCP ACK - Transaction ID 0xfe2a0d38
3	5.251945872	PCSSystemtec_78:91:...	PCSSystemtec_00:86:...	ARP	42	Who has 10.0.2.2? Tell 10.0.2.19
4	5.252347104	PCSSystemtec_00:86:...	PCSSystemtec_78:91:...	ARP	60	10.0.2.2 is at 08:00:27:00:86:d7
5	11.152816540	PCSSystemtec_78:91:...	Broadcast	ARP	42	Who has 10.0.2.15? Tell 10.0.2.19
6	11.153363556	PCSSystemtec_c1:5f:...	PCSSystemtec_78:91:...	ARP	60	10.0.2.15 is at 08:00:27:c1:5f:83
7	11.260451014	10.0.2.19	192.168.0.1	DNS	82	Standard query 0x44ee PTR 15.2.0.10.in-addr.arpa
8	11.272557402	192.168.0.1	10.0.2.19	DNS	105	Standard query response 0x44ee PTR 15.2.0.10.in-addr.arpa PTR 10.0.2.19
9	11.284856187	10.0.2.19	10.0.2.15	TCP	54	44904 → 443 [FIN] Seq=1 Win=1024 Len=0
10	11.284922762	10.0.2.19	10.0.2.15	TCP	54	44904 → 22 [FIN] Seq=1 Win=1024 Len=0
11	12.387020387	10.0.2.19	10.0.2.15	TCP	54	44906 → 22 [FIN] Seq=1 Win=1024 Len=0
12	12.387083395	10.0.2.19	10.0.2.15	TCP	54	44906 → 443 [FIN] Seq=1 Win=1024 Len=0
13	16.516217354	PCSSystemtec_78:91:...	52:55:0a:00:02:01	ARP	42	Who has 10.0.2.1? Tell 10.0.2.19
14	16.516719844	52:55:0a:00:02:01	PCSSystemtec_78:91:...	ARP	60	10.0.2.1 is at 52:55:0a:00:02:01

Explanation:

Inverse scanning sends TCP FIN packets instead of SYN packets. The behavior observed in Wireshark:

- If a port is closed: The target responds with a TCP RST.
- If a port is open or filtered: The target ignores the FIN packet (no response).

Wireshark observation:

For both Metasploitable 2 and Windows 10, no RST packets were received for ports 22 and 443, leading to the open|filtered state. This indicates either the ports are open but not responding to FIN, or they are filtered by a firewall.

c. Conduct UDP scanning on at least one open UDP port and one closed UDP port.

i. Record your result in the table below.

Host	Port Number (UDP)	Status (Open/Closed)
Metasploitable 2	53	Open
	80	Closed
Windows 10	53	Open filtered
	80	Open filtered

Metasploitable 2

```
(pentest@pentest) ~$ sudo nmap -sU -p 53,80 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 12:49 +08
Nmap scan report for 10.0.2.3 (10.0.2.3)
Host is up (0.00051s latency).
PORT      STATE SERVICE
53/udp    open  domain
80/udp    closed http
MAC Address: 08:00:27:BF:43:AD (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds
```

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Windows 10

```
(pentest@PenTest)-[~]
$ sudo nmap -sU -p 53,80 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 12:54 +08
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00070s latency).

```

PORT	STATE	SERVICE
53/udp	open filtered	domain
80/udp	open filtered	http

```
MAC Address: 08:00:27:C1:5F:83 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 1.74 seconds
```

- ii. Use the Wireshark tool to study how NMAP conducts UDP scan.

Screenshot the Wireshark output and **explain** how the UDP scanning works based on the Wireshark output.

Metasploitable 2

No.	Time	Source	Destination	Protocol	Length	Info
7	102.197768051	192.168.0.1	10.0.2.19	DNS	103	Standard query response 0x6640 PTR 3.2.0.10.in-addr.arpa PTR 10.0.2.19
8	102.514657291	10.0.2.19	10.0.2.3	DNS	72	Standard query 0x0006 TXT version.bind
9	102.514715840	10.0.2.19	10.0.2.3	DNS	54	Server status request 0x0000
10	102.514732860	10.0.2.19	10.0.2.3	DNS	88	Standard query 0x0000 PTR _services._dns-sd._udp.local
11	102.514757078	10.0.2.19	10.0.2.3	UDP	56	60002 → 80 Len=14
12	102.515147425	10.0.2.3	10.0.2.19	ICMP	84	Destination unreachable (Port unreachable)
13	102.515535176	10.0.2.3	10.0.2.19	DNS	104	Standard query response 0x0006 TXT version.bind TXT NS version.bind
14	102.515575632	10.0.2.19	10.0.2.3	ICMP	132	Destination unreachable (Port unreachable)
15	102.515707054	10.0.2.3	10.0.2.19	DNS	60	Server status request response 0x0000 Not implemented
16	102.515746097	10.0.2.19	10.0.2.3	ICMP	82	Destination unreachable (Port unreachable)
17	102.515911068	10.0.2.3	10.0.2.19	DNS	299	Standard query response 0x0000 PTR _services._dns-sd._udp.local NS
18	102.515943125	10.0.2.19	10.0.2.3	ICMP	327	Destination unreachable (Port unreachable)
19	107.369896586	PCSSystemtec_78:91:...	52:55:0a:00:02:01	ARP	42	Who has 10.0.2.1? Tell 10.0.2.19

Windows 10

No.	Time	Source	Destination	Protocol	Length	Info
69	151.517295058	192.168.0.1	10.0.2.19	DNS	105	Standard query response 0x72a9 PTR 15.2.0.10.in-addr.arpa PTR 10.0.2.19
70	151.781674631	10.0.2.19	10.0.2.15	UDP	56	33407 → 80 Len=14
71	151.781738042	10.0.2.19	10.0.2.15	DNS	72	Standard query 0x0006 TXT version.bind
72	151.781754170	10.0.2.19	10.0.2.15	DNS	54	Server status request 0x0000
73	151.781771030	10.0.2.19	10.0.2.15	DNS	88	Standard query 0x0000 PTR _services._dns-sd._udp.local
74	152.883111629	10.0.2.19	10.0.2.15	DNS	72	Standard query 0x0006 TXT version.bind
75	152.883179670	10.0.2.19	10.0.2.15	DNS	54	Server status request 0x0000
76	152.883202424	10.0.2.19	10.0.2.15	DNS	88	Standard query 0x0000 PTR _services._dns-sd._udp.local
77	152.883227604	10.0.2.19	10.0.2.15	UDP	56	33409 → 80 Len=14
78	156.516631495	PCSSystemtec_78:91:...	52:55:0a:00:02:01	ARP	42	Who has 10.0.2.1? Tell 10.0.2.19
79	156.517131453	52:55:0a:00:02:01	PCSSystemtec_78:91:...	ARP	64	10.0.2.1 is at 52:55:0a:00:02:01
80	178.236511167	fe80::30b0:1664:772...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
81	178.719740083	fe80::30b0:1664:772...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2

Explanation:

UDP scanning sends UDP datagrams to target ports:

- If a port is open: The service may reply with a UDP packet (e.g., DNS on port 53). If not, Nmap may mark it as open|filtered.
- If a port is closed: The target sends an ICMP Port Unreachable message.

Wireshark observation:

- For Metasploitable 2 (port 53), DNS queries were sent and responses received → Open.
- For Metasploitable 2 (port 80), no response → Closed (or ICMP unreachable was seen).

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- For Windows 10, no UDP responses were received for ports 53 or 80 → open|filtered (likely firewall dropping UDP packets).

5. Once the hosts is identified, the next step is to find out what is the OS that a particular host runs so that their vulnerability can be narrowed down. However, OS tend to make it difficult to accurately determine their OS. Therefore, OS fingerprinting is conducted.

a. Conduct **active** OS fingerprinting using NMAP tool to discover the OS that each host runs.

i. Record your results in the table below.

Host	OS Detected	True OS discovered? (Yes/No)
Metasploitable 2	Linux 2.6.9 – Linux 2.6.33	Yes
Windows 10	Too many fingerprints matched	No

Metasploitable 2

```
(pentest@PenTest)-[~]
$ sudo nmap -o 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 13:04 +08
Nmap scan report for 10.0.2.3 (10.0.2.3)
Host is up (0.00058s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
```

LAB 3: SCANNING AND ENUMERATION

```
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:BF:43:AD (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.95 seconds
```

Windows 10

```
(pentest@PenTest)-[~]
$ sudo nmap -O 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 13:06 +08
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00071s latency).
All 1000 scanned ports on 10.0.2.15 (10.0.2.15) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 08:00:27:C1:5F:83 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.21 seconds
```

- ii. Did all the true OS was discovered? Explain your findings.

Metasploitable 2:

Correctly identified as Linux 2.6.9–2.6.33. This is accurate because Metasploitable 2 runs an older Ubuntu with a 2.6.24 kernel, which falls within this range.

Windows 10:

Nmap could not definitively identify the OS because all scanned ports were filtered. Without open ports to probe for OS-specific TCP/IP stack behaviors, Nmap lacks sufficient data to make a precise match.

Not all true OSes were discovered. Windows 10 remained undetected due to its restrictive firewall.

LAB 3: SCANNING AND ENUMERATION

- b. Conduct **passive** OS fingerprinting using P0f tool to discover the OS in the VM NAT network.
 - i. Explain the steps you take for **passive** fingerprinting.
Run `sudo p0f -i eth0` to listen on the network interface, then generate traffic (SSH attempt from Windows 10 to Kali) to capture OS signatures.

```
(pentest@PenTest)-[~]
$ sudo p0f -i eth0
p0f 3.09b by Michal Zalewski <lcamtuf@coredump.cx>

[+] Closed 1 file descriptor.
[+] Loaded 322 signatures from '/etc/p0f/p0f.fp'.
[+] Intercepting traffic on interface 'eth0'.
[+] Default packet filtering configured [+VLAN].
[+] Entered main event loop.
```

```
C:\Users\IEUser>ssh 10.0.2.19
ssh: connect to host 10.0.2.19 port 22: Connection refused
```

```

-[-[ 10.0.2.15/49796 → 10.0.2.19/22 (syn) ]-]
|
| client      = 10.0.2.15/49796
| os          = Windows NT kernel
| dist       = 0
| params     = generic
| raw_sig    = 4:128+0:0:1460:mss*44,8:mss,nop,ws,nop,nop,sok:df,id+:0
|
| _____
|
| client      = 10.0.2.15/49796
| link       = Ethernet or modem
| raw_mtu    = 1500
|
| _____
|
-[-[ 10.0.2.15/49796 → 10.0.2.19/22 (syn) ]-]
|
| client      = 10.0.2.15/49796
| os          = Windows NT kernel
| dist       = 0
| params     = generic
| raw_sig    = 4:128+0:0:1460:mss*44,8:mss,nop,ws,nop,nop,sok:df,id+:0
|
| _____

```

LAB 3: SCANNING AND ENUMERATION

```

.-[ 10.0.2.15/49796 → 10.0.2.19/22 (mtu) ]-
|
| client      = 10.0.2.15/49796
| link       = Ethernet or modem
| raw_mtu    = 1500
|
|-----|
| 08:00:1864:7720:FB2::1:2 | ICMPv6 62 Router Solicitation
| 08:00:1864:7720:FB2::1:2 | ICMPv6 150 Multicast Listener Report Message v2
| 08:00:1864:7720:FB2::1:2 | ICMPv6 62 Membership Report / Join group 224.0.0.251 ff02::c
| 08:00:1864:7720:FB2::1:2 | ICMPv6 130 Multicast Listener Report Message v2
| 08:00:1864:7720:FB2::1:2 | DHCPv6 153 Solicit XID: 8x7e2fd8 CID: 000100012423100500
| 08:00:1864:7720:FB2::1:2 | ARP 60 Who has 10.0.2.15? (ARP Probe)
| 08:00:1864:7720:FB2::1:2 | ICMPv6 86 Neighbor Advertisement fe80::3608:1664:7720::6
| 08:00:1864:7720:FB2::1:2 | DHCPv6 153 Solicit XID: 8x7e2fd8 CID: 000100012423100500
| 08:00:1864:7720:FB2::1:2 | ICMPv6 60 Who has 10.0.2.15? (ARP Probe)
.-[ 10.0.2.15/49796 → 10.0.2.19/22 (syn) ]-
| 08:00:1864:7720:FB2::1:2 | ICMPv6 70 Router Solicitation from 00:08:27:c1:5f:83
| 08:00:1864:7720:FB2::1:2 | DHCPv6 153 Solicit XID: 8x7e2fd8 CID: 000100012423100500
| 08:00:1864:7720:FB2::1:2 | ICMPv6 70 Router Solicitation from 00:08:27:c1:5f:83
|
| client      = 10.0.2.15/49796
| os         = Windows NT kernel
| dist       = 0
| params     = generic
| raw_sig    = 4:128+0:0:1460:mss*44,8:mss,nop,ws,nop,nop,sok:df,id+:0
|
|-----|
.-[ 10.0.2.15/49796 → 10.0.2.19/22 (mtu) ]-
|
| client      = 10.0.2.15/49796
| link       = Ethernet or modem
| raw_mtu    = 1500
|

```

- ii. Record your result in the table below.

Host	OS(s) Detected	True OS discovered? (Yes/No)
Metasploitable 2	Linux	Yes
Windows 10	Windows NT kernel	Yes

6. Besides OS fingerprinting, banner grabbing (using -A in nmap) can also be useful to discover the host's OS and services.
 - a. Run banner grabbing using NMAP on all the VM hosts and record the result in the table.

LAB 3: SCANNING AND ENUMERATION

Metasploitable 2

```
(pentest@PenTest)-[~]
$ sudo nmap -A 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 13:17 +08
Nmap scan report for 10.0.2.3 (10.0.2.3)
Host is up (0.00059s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst: 1571 Broadcast ARP 68 ARP Announcement for 10.0.2.15
|_STAT: 684:772 ff02::2 ICMPv6 68 ARP Solicitation from 08:00:27:c1:5f:83
|_FTP server status: 2 ICMPv6 78 Router Solicitation from 08:00:27:c1:5f:83
|_Connected to 10.0.2.19
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-date: 2025-12-01T05:18:18+00:00; 0s from scanner time.
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/st
ateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_SSL2_RC4_128_WITH_MD5
|_SSL2_RC4_128_EXPORT40_WITH_MD5
|_SSL2_DES_192_EDE3_CBC_WITH_MD5
|_SSL2_RC2_128_CBC_WITH_MD5
|_SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_SSL2_DES_64_CBC_WITH_MD5
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_program version port/proto service
|_100000 2 111/tcp rpcbind
|_100000 2 111/udp rpcbind
|_100003 2,3,4 2049/tcp nfs
|_100003 2,3,4 2049/udp nfs
|_100005 1,2,3 37875/udp mountd
|_100005 1,2,3 40316/tcp mountd
```

LAB 3: SCANNING AND ENUMERATION

```
| 100005 1,2,3 40316/tcp mountd
| 100021 1,3,4 35540/tcp nlockmgr
| 100021 1,3,4 39067/udp nlockmgr
| 100024 1 52484/udp status
| 100024 1 54054/tcp status
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open exec?
513/tcp open login OpenBSD or Solaris rlogind
514/tcp open tcpwrapped
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
| mysql-info:
| Protocol: 10
| Version: 5.0.51a-3ubuntu5
| Thread ID: 8
| Capabilities flags: 43564
| Some Capabilities: SupportsCompression, Support41Auth, SupportsTransactions, LongColumnFlag, Speaks41ProtocolNew, SwitchToSSLAfterHandshake, ConnectWithDatabase
| Status: Autocommit
| Salt: s[NVF9+iR>~%5}66KTGG
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/st
ateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
| Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2025-12-01T05:18:18+00:00; 0s from scanner time.
5900/tcp open vnc VNC (protocol 3.3)
| vnc-info:
| Protocol version: 3.3
| Security types:
|_ VNC Authentication (2)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
| irc-info:
| users: 1
| servers: 1
| lusers: 1
| lservers: 0
| server: irc.Metasploitable.LAN
| version: Unreal3.2.8.1. irc.Metasploitable.LAN
| uptime: 0 days, 1:35:55
| source ident: nmap
| source host: AC02C333.EB72D3BE.7B559A54.IP
|_ error: Closing Link: kfuzhpsna[10.0.2.19] (Quit: kfuzhpsna)
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
|_http-title: Apache Tomcat/5.5
|_http-favicon: Apache Tomcat
|_http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:BF:43:AD (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```


LAB 3: SCANNING AND ENUMERATION

```
MAC Address: 08:00:27:BF:43:AD (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix,
Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2025-12-01T00:18:10-05:00
|_ clock-skew: mean: 1h15m00s, deviation: 2h30m01s, median: 0s
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ smb2-time: Protocol negotiation failed (SMB2)

TRACEROUTE
HOP RTT ADDRESS
1 0.58 ms 10.0.2.3 (10.0.2.3)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 76.27 seconds
```

Windows 10

```
(pentest@PenTest)-[~]
$ sudo nmap -A 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-01 13:23 +08
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00063s latency).
All 1000 scanned ports on 10.0.2.15 (10.0.2.15) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 08:00:27:C1:5F:83 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

TRACEROUTE
HOP RTT ADDRESS
1 0.63 ms 10.0.2.15 (10.0.2.15)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 30.00 seconds
```

LAB 3: SCANNING AND ENUMERATION

- b. Record your result in the table below.

Host	OS(s) Detected	True OS discovered? (Yes/No)	Services offered	Service provider and version
Metasploitable 2	Linux 2.6.9 – Linux 2.6.33	Yes	ftp	Vsftpd 2.3.4
			ssh	OpenSSH 4.7p1 Debian 8ubuntu1
			telnet	Linux tenetd
			smtp	Postfix smtpd
			dns	ISC Bind 9.4.2
			http	Apache/2.2.8 (Ubuntu)
			RPCBind	Rcpbind v2
			NETBIOS	Samba smbd (3.0.20-Debian)
			MySQL	5.0.51a-3ubuntu5
			PostgreSQL	8.3.0-8.3.7
			VNC	VNC(protocol 3.3)
			IRC	UneallIRCd 3.2.8.1
			Tomcat	Apache Tomcat/Coyote JSP engine 1.1
Windows 10	Not specified	No	None	No open service detected. All ports were filtered

LAB 3: SCANNING AND ENUMERATION

7. Use the Zenmap software to map the VM NAT network. Screenshot your result below.

